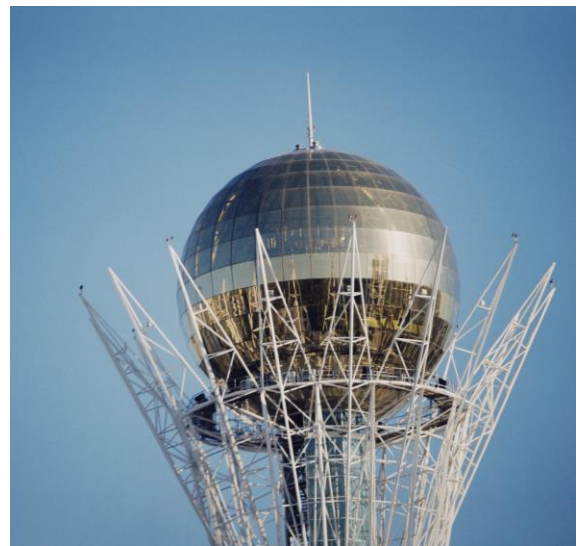
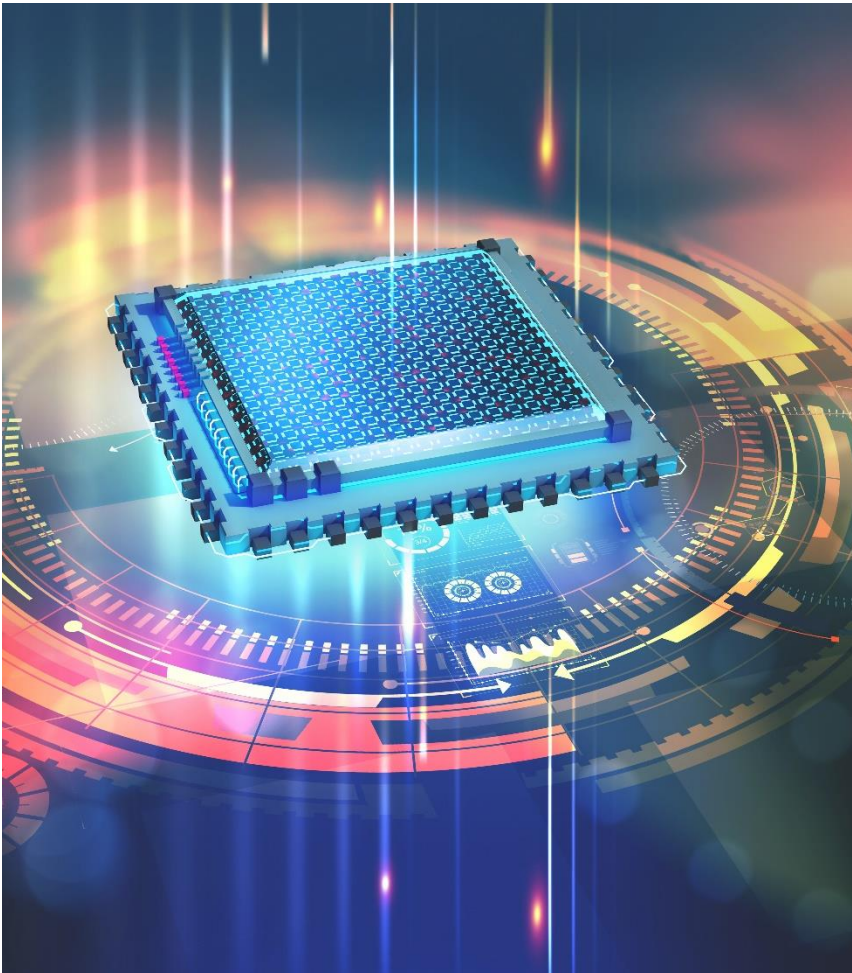


Model 805-SG

Ultra-Agile Microwave Signal Source



Features

- Microsecond switching-speed
- Low-phase noise and spur levels
- Portable battery option
- Low level error accuracy

Applications

- QuickSyn FSW-0010/0020 replacement
- Automated Test Environment
- Characterizing antennas, semiconductor devices, and other components
- Radar signal generation and Electronic warfare



Model 805-SG Datasheet
8 kHz to 20 GHz

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DEFINITIONS

The specifications in the following pages describe the warranted performance of the instrument for 23 ± 5 °C after a 30-minute warm-up period (unless otherwise stated).

Min/Max: Parameter range that is guaranteed by product design, and/or production tested. Warranted performance specifications include guard-bands to account for the expected statistical performance distribution, measurement uncertainties, and changes in performance due to environmental conditions.

Typical: Expected mean values, not warranted performance.

INTRODUCTION

The **Model 805-SG** microwave signal source modules deliver instrument-grade performance, increased functionality, and efficient power consumption at a reduced size and affordable cost. The design combines low phase noise with fast switching capability, covering a wide frequency range from 8 kHz up to 20 GHz. The low spurious and harmonic content of the signal makes it ideally suitable for many demanding applications.

The unit contains a high stability OCXO, providing accurate, power-calibrated, phase-lockable output signals.

The frequency resolution is 1 mHz and the power resolution is 0.01 dB power.

The unit is remotely controlled with USB, LAN or SPI control.

Due to the form-factor, the unit can also be used as a drop-in replacement of the obsolete “QuickSyn Synthesizers” from NI.

SPECIFICATIONS

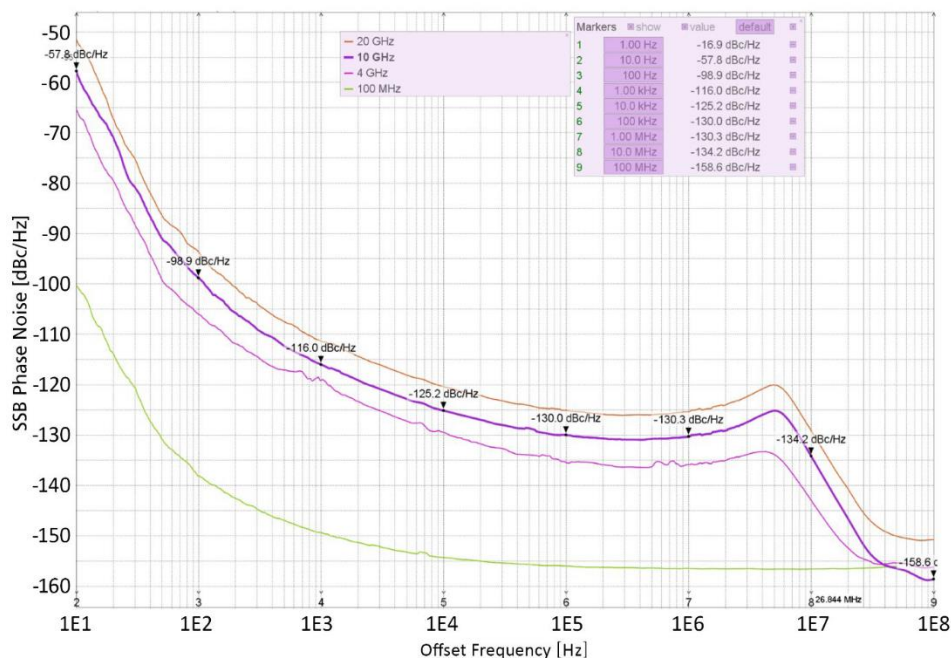
Signal Specifications

PARAMETER	MIN	TYPICAL	MAX	NOTE
Frequency Range	10 MHz 8 kHz		20 GHz 20 GHz	settable to 22 GHz Option 8K
Frequency Resolution		0.001 Hz 0.001 Hz		GUI SW setting resolution SPI interface setting resolution
Frequency switching time		500 μ s 15 μ s	20 μ s	Option FS
Phase adjustment range	0 deg		360 deg	
Phase resolution		0.1 deg		

Phase Noise

OFFSET	10 Hz		100 Hz		1 kHz		20 kHz		100 kHz		1 MHz		10 MHz	
FREQUENCY	Typ.	Max.	Typ.	Max.	Typ.	Max.	Typ.	Max.	Typ.	Max.	Typ.	Max.	Typ.	Max.
100 MHz	-108	-103	-136	-131	-150	-145	-155	-150	-155	-150	-155	-150	-154	-149
1 GHz	-80	-75	-116	-111	-137	-132	-147	-142	-147	-142	-145	-140	-156	-151
2 GHz	-74	-69	-112	-107	-133	-128	-144	-139	-144	-139	-141	-136	-155	-150
5 GHz	-65	-60	-101	-96	-126	-121	-136	-131	-138	-133	-136	-131	-148	-143
10 GHz	-60	-55	-96	-91	-120	-115	-130	-125	-131	-126	-126	-124	-142	-137
20 GHz	-52	-47	-91	-86	-114	-109	-125	-120	-126	-121	-124	-119	-137	-132

Figure 1: SSB Phase Noise Performance



Spectral Purity

PARAMETER	MIN	TYPICAL	MAX	NOTE
Harmonics				At 10 dBm; See plot below
< 100 MHz		-35 dBc		
100 MHz to 20 GHz		-50 dBc	-40 dBc	
4.5 GHz to 20 GHz		-50 dBc	-40 dBc	
Sub-harmonics				At 10 dBm; See plot below
< 5 GHz		-75 dBc	-65 dBc	

5 GHz to 11.5 GHz		-65 dBc	-55 dBc	
11.5 GHz to 20 GHz		-75 dBc	-65 dBc	
Non-harmonic spurious				10 kHz to 0.5 GHz offset from carrier
< 1 GHz		-65 dBc	-55 dBc	
1 GHz to 20 GHz		-70 dBc	-60 dBc	

Figure 2: Harmonics (at 10 dBm Output Power)

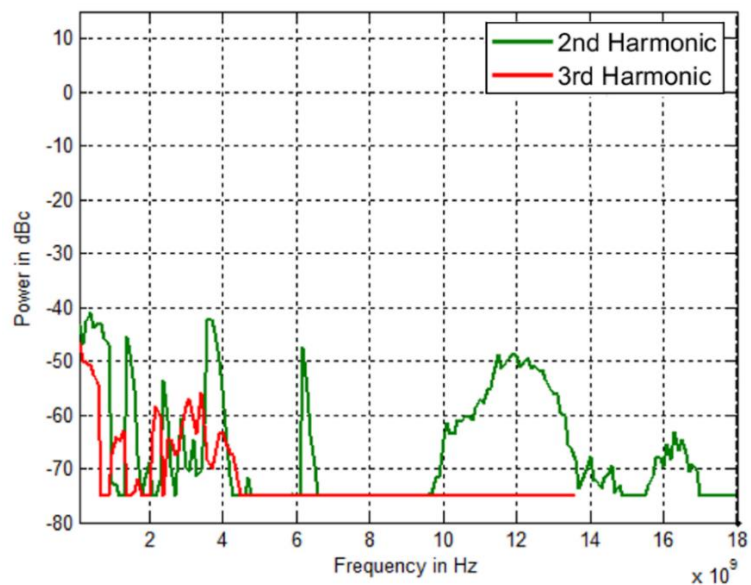


Figure 3: Wideband Spectrum for 5 GHz

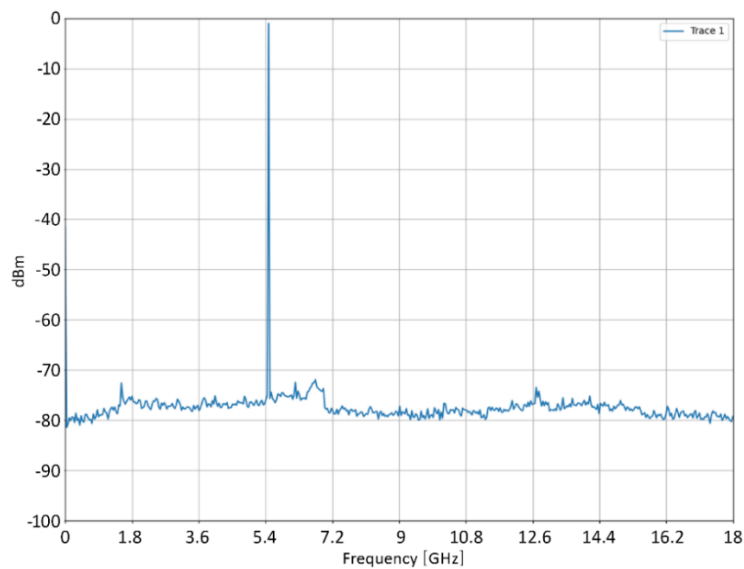
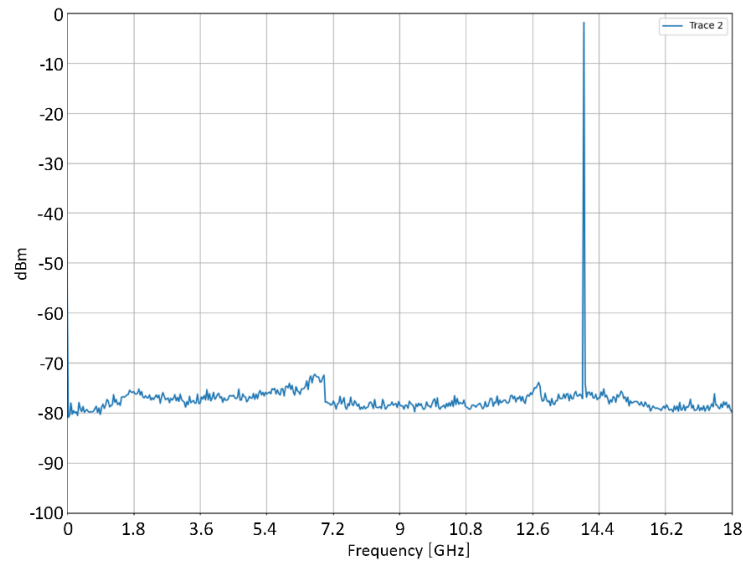


Figure 4: Wideband Spectrum for 14 GHz



Level Performance

PARAMETER	MIN	TYPICAL	MAX	NOTE
Output power level				Settable to +25 dBm; See plot below
8 kHz to 10 MHz	-20 dBm		9 dBm	
10 GHz to 20 GHz	-20 dBm		15 dBm	
Power level uncertainty		0.25 dB	1.0 dB	-20 to 15 dBm See plots below
Power resolution		0.01 dB 0.1 dB		GUI SW setting resolution SPI interface setting resolution
Power settling time		1 ms	5 μ s	ALC off (open loop) ALC on (closed loop)
Output impedance		50 Ω		
VSWR		1.7		
Reverse Power protection				
DC voltage			7 V	
RF power			23 dBm	

Figure 5: Power level linearity

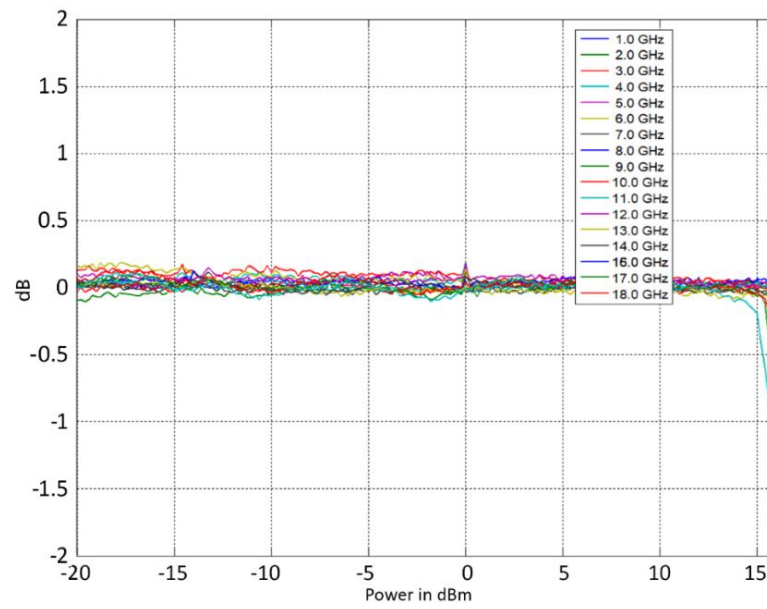
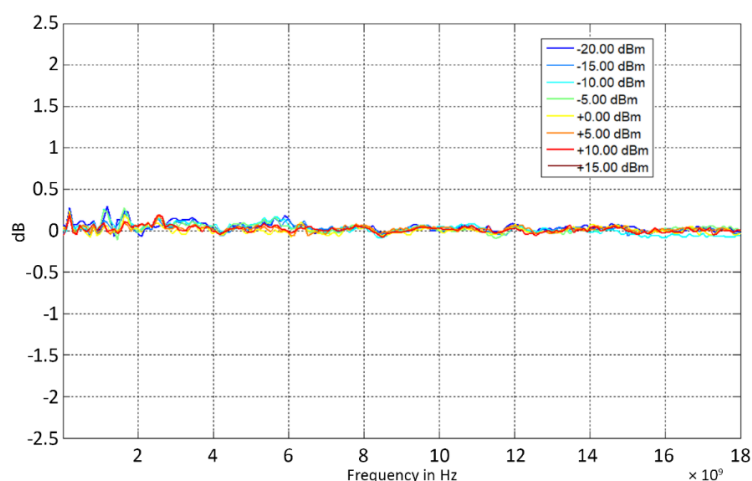


Figure 6: Frequency Response



Reference Frequency

PARAMETER	MIN	TYPICAL	MAX	NOTE
Internal reference frequency		100 MHz		
Calibrated accuracy of int. reference		±30 ppb		Calibrated at 23 ± 3 °C
Temperature stability (0 to 50 °C)			±100 ppb	
Aging 1 st year			500 ppb	
Aging per day			5 ppb	After 30 days operation
Warm-Up Time		5 min		
Reference frequency input	10 MHz, 100 MHz, 1 GHz 10 – 250 MHz			Fixed reference frequency Variable reference frequency
Reference input level				
10 MHz	-3 dBm		+12 dBm	Fixed reference frequency
100 MHz	-3 dBm		+12 dBm	Fixed reference frequency
1 GHz	-3 dBm		+12 dBm	Fixed reference frequency
10 – 250 MHz	-3 dBm		+12 dBm	Variable reference frequency
Variable reference frequency resolution		100 kHz		
Lock range				
10 MHz			±1.5 ppm	Fixed reference frequency
100 MHz			±100 ppm	Fixed reference frequency, bypass
1 GHz			±100 ppm	Fixed reference frequency, bypass
10 – 250 MHz			±1.5 ppm	Variable reference frequency
Reference input impedance		50 Ω		
Reference frequency output	10 MHz, 100 MHz, 1 GHz			Settable
Output power				
10 MHz	+3 dBm	+5 dBm	+7 dBm	
100 MHz	+3 dBm	+5 dBm	+7 dBm	
1 GHz	+3 dBm	+5 dBm	+7 dBm	
Reference output impedance		50 Ω		

Reference architecture

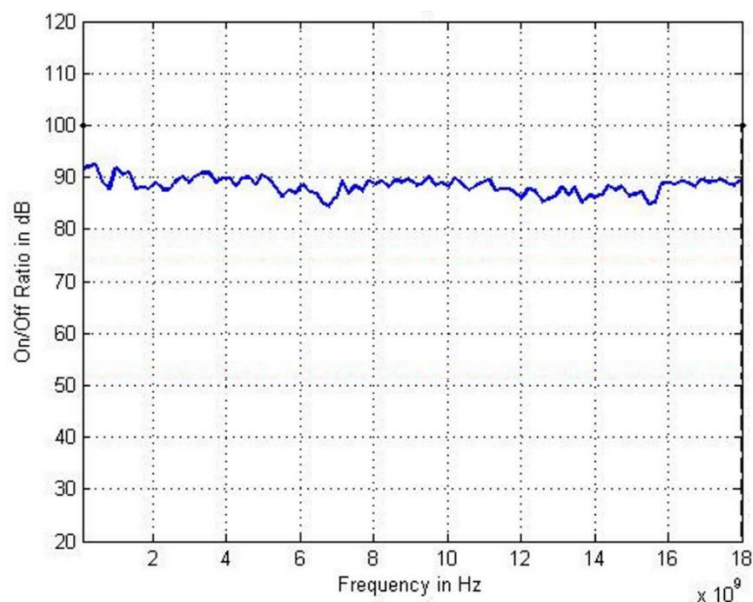
External variable reference frequencies and 10 MHz external fixed reference frequency will be internally locked to the internal 100 MHz reference with a PLL circuit. 100 MHz and 1 GHz external fixed reference frequencies are bypassing the internal reference PLL circuit and are acting directly as reference signal for the synthesizer.

Modulation Capability

PARAMETER	MIN	TYPICAL	MAX	NOTE
Pulse Modulation				Option PULSE
Modulation source	Internal			

	External (PULSE)			
On/off ratio				At 10 dBm; see plot below.
< 1 GHz	70 dB	85 dB		
1 GHz to 20 GHz	80 dB	100 dB		
Repetition frequency	0 Hz		10 MHz	
Pulse width	30 ns		20 s	
Pulse width compression		20 ns		
Pulse resolution		10 ns		
Pulse rise/fall time		9 ns	12 ns	
Pulse polarity		Normal Inverse		Settable
Pulse train length	1		4096	
Pulse overshoot			10%	
External pulse latency		45 ns	60 ns	

Figure 7: On/Off Ratio – pulse modulation (at 10 dBm Output Power)



Sweeping Capability

PARAMETER	MIN	TYPICAL	MAX	NOTE
Sweep parameters	Frequency, power, list			
Number of list points	1		50,000	
Sweep type	Linear, logarithmic, random			
Step time	500 μ s 20 μ s		20 s 20 s	Option FS
Timing Resolution		10 ns		
Timing accuracy per point		20 ns		

Generalized list sweep

Allows for individual setting of frequency, step-time, and off-time for each point

Trigger (PULSE)

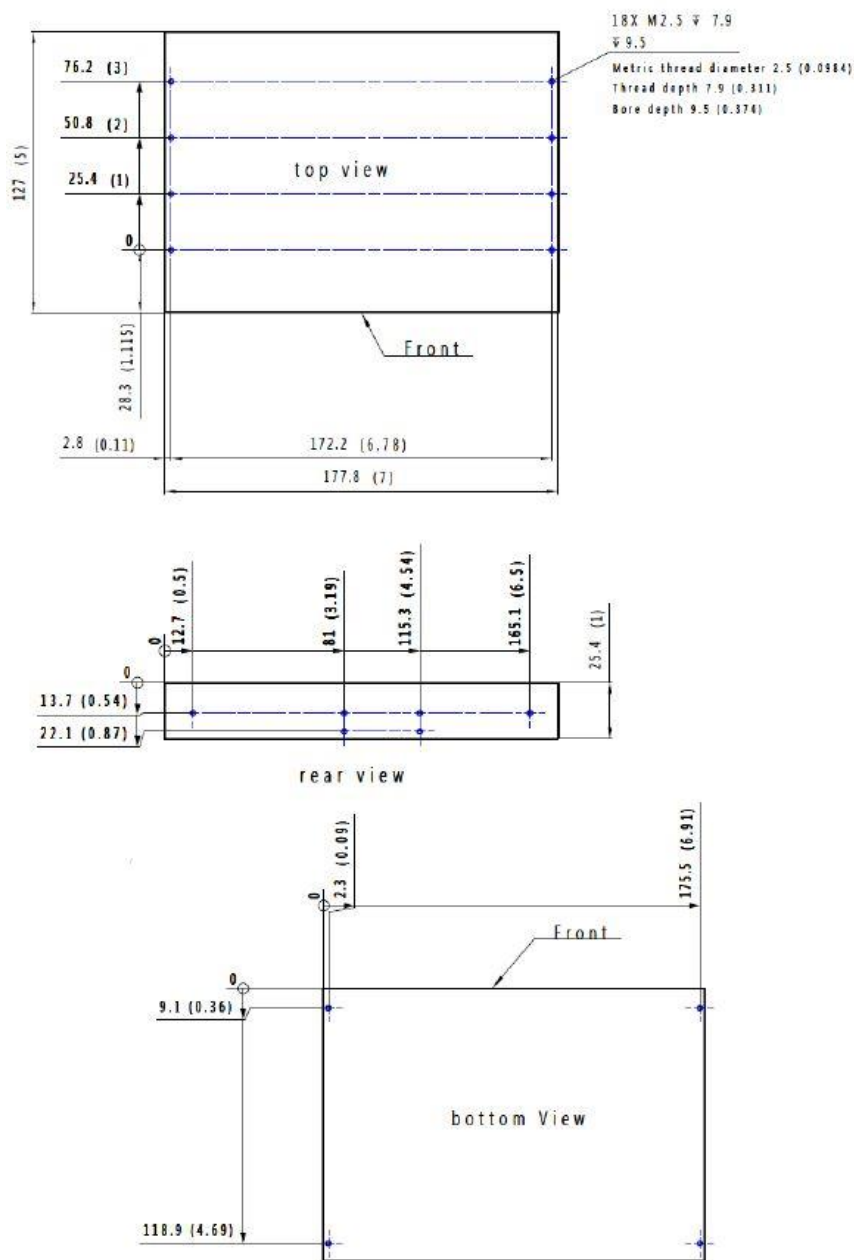
PARAMETER	MIN	TYPICAL	MAX	NOTE
Trigger types		Continuous Single (point) Gated		

	Gated direction			
Trigger source	External (PULSE, SPI Trigger) Bus (Ethernet, USB, SPI)			
Trigger Modes	Continuous free run Trigger and run Reset and run			
External trigger latency		140 ns		
External trigger uncertainty		20 ns		
External trigger delay	0 s		20 s	Settable
External delay resolution		10 ns		
Trigger modulo	1		255	Execute only on Nth trigger event
Trigger polarity		Rising Falling		
External PULSE input threshold	0.85 V	0.9 V	0.95 V	TTL compatible
External PULSE input voltage range	-0.5 V		+5.5 V	TTL compatible 100 k Ω pull-up to +5.0 V
External PULSE input hysteresis		30 mV		

MECHANICAL SPECIFICATIONS

Dimensions & Weight

PARAMETER	VALUE
Including connectors	W x L x H = 7 x 5.4 x 1 in
Excluding connectors	W x L x H = 7 x 5 x 1 in
Weight	< 2.4 lb



Installation Instructions

The module relies on passive cooling. It is mandatory to mount the device on a heatsinking surface. Make sure the synthesizer operates under the conditions specified in this datasheet. Otherwise, the thermal protection will turn off the RF output.

INTERFACES

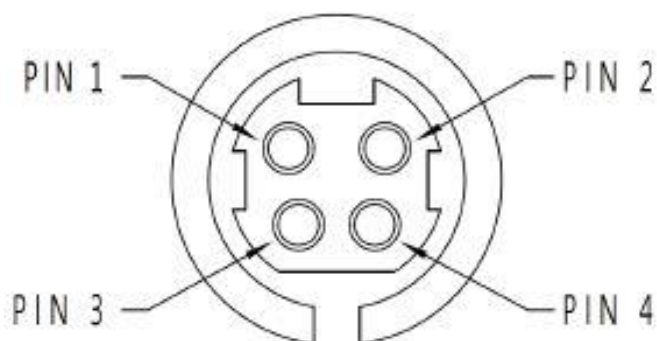
Front Panel

LABEL	TYPE	DESCRIPTION
1. DC IN	KPIX-4S	DC input (see also chapter “Power Connector Assembly”). Redundant power supply input to the SPI Interface DC input (supply with higher voltage will be chosen).
2. SPI	DF1BZ-20DP-2.5DS	SPI Interface, including DC input (see also chapter “SPI Interface”)
3. ETH	RJ-45	Ethernet port
4. USB	Micro B	USB Port
5. PULSE	SMA	Trigger / Pulse interface, 100 k Ω pull-up to +5.0 V
6. PWR	LED	Power ON/OFF indicator
7. REM	LED	Remote connection status indicator
8. RF	LED	RF output ON/OFF indicator
9. REF OUT	SMA	Reference signal output
10. REF IN	SMA	Reference signal input
11. RF OUT	SMA	RF output

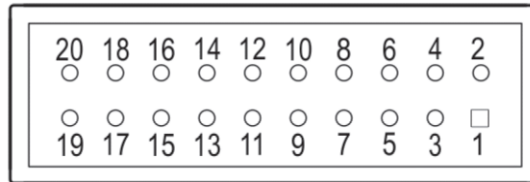


Power Connector Assembly

PIN	ASSIGNMENT
1	GND
2	DC Supply (see also “Power requirements”)
3	GND
4	DC Supply (see also “Power requirements”)

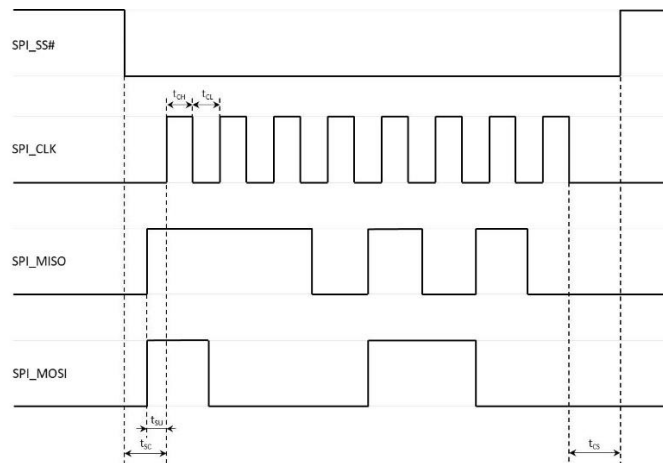


The power connector is a 4 pin, snap and lock receptacle. BNC recommends Kycon manufactured plugs KPPX-4P from its KPPX series.



SIGNAL	PIN	TYPE	DESCRIPTION
SPI_CLK	11	Input	SPI clock. Supplied by the controlling host. The controlling host is the SPI master, the synthesizer is the SPI slave.
SPI_SS#	13	Input	SPI Slave Select. This signal is an active low input from the host to the synthesizer. It frames command communications. For each command, SPI_SS# goes low before the first bit is sent and goes high after the last bit is sent
SPI_MISO	7	Output	Master In/Slave Out. Data line from the synthesizer to the host.
SPI_MOSI	9	Input	Master Out/Slave In. Command/Data line from the host to the synthesizer.
TRIGGER	17	Input	Edge sensitive input. The trigger signal of +3.3 V can be configured for multiple trigger modes (see also chapter “Trigger (PULSE)”).
LOCK	15	Output	Output indicates the RF output of the synthesizer is locked on its current setting (+3.3 V locked, 0 V unlocked).
REF_LOCK	16	Output	Output indicates the synthesizer has detected an external reference signal and locked on that signal (+3.3 V locked, 0 V unlocked).
RESET#	18	Input	Internally pulled up to +3.3 V with 100 kΩ resistor. Active low signal, which has a minimum width of 1 ms, will reset the synthesizer to a default state.
DC IN	3, 4	Input	External power supply (see also “Power requirements”). Redundant power supply input to the DC IN interface (supply with higher voltage will be chosen).
GND	8, 10, 19, 20		Ground.
DNC	1, 2, 5, 6, 12, 14		Do not connect. Reserved for factory / future use.

The SPI interface connector is a 20 pin, 2.50 mm spaced double-row header. BNC recommends HIROSE manufactured socket DF1B-20DS-2.5RC and corresponding contacts from its DF1B series.



t_{SC}	> 25 ns	SPI_SS# to be low before first clock edge
t_{CS}	> 25 ns	SPI_CLK to be low before releasing SPI_SS#
t_{SU}	> 15 ns	SPI_MISO/MOSI to be stable before rising edge of clock
t_{CH}	> 25 ns	Minimum high time of a clock pulse
t_{CL}	> 25 ns	Minimum low time of a clock pulse
f_{CLK}	≤ 12 MHz	Maximum clock frequency

ORDERING INFORMATION

HOST MODEL	PRODUCT	DESCRIPTION
805-SG	805-SG-1	20 GHz wideband frequency synthesizer module (with AC adapter)
805-SG	Option FS	Fast switching
805-SG	Option 8K	Frequency Range extension to 8 kHz
805-SG	Option PULSE	Internal / external pulse modulation

GENERAL CHARACTERISTICS

Remote Programming Interfaces:

Ethernet

USB 2.0

SPI

Control language: SCPI Version 1999.0, native command set.

DC Power requirements: 12.0 – 30.0 V_{DC}; 24 W typical

AC Mains Adapter (supplied): 100-240 VAC in / 24 V, 2.7 A DC out

Storage temperature range: -40 to 70 °C

Operating temperature range: 0 to 60 °C, non-condensing, temperature of passive heatsink

Operating and storage altitude: up to 15,000 feet



Safety/EMC complies with applicable Safety and EMC regulations and directives.

Recommended calibration cycle: 24 months

Document History

Version	Date	Author	Notes
V100	2023-6-19	Dd	First draft of the datasheet
V101	2023-8-10	Jk	Added product picture, refine specs
V102	2023-12-01	Ap	Added modulations, redefined operating temperature range
V103	2024-2-12	AT	Removed option MOD, updated reference frequency, modulation capability, and SPI interface
V104	2024-06-14	dd/ys/ap	Refine specification, added new plot maximum power 8 kHz to 500 kHz

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NOTES

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